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A hybrid ensemble stacking and explainable machine learning framework enhanced with bald eagle search optimization for coronary artery disease risk prediction

Komal Kumar Napa[1*], Sangeetha Murugan[2], Senthil Murugan J[3], S. Sathya[4], A. G. Balamurugan[5] and Nageswari D[6]

*Correspondence:
Komal Kumar Napa
komalkumarnapa@gmail.com
[1]Department of Artificial Intelligence and Data Science, Saveetha Engineering College, Chennai, Tamil Nadu, India
[2]Department of Computer Science & Engineering, Madanapalle Institute of Technology & Science (MITS), Deemed to be University, Madanapalle, Andhra Pradesh, India
[3]Department of Computer Science and Engineering, Vel Tech High Tech Dr. Rangarajan Dr. Sakunthala Engineering College, Chennai, India
[4]Department of Electronics and Communication Engineering, S.A. Engineering College, Chennai, India
[5]Department of Computer Science & Engineering, Vel Tech Rangarajan Dr.Sagunthala R&D Institute of Science and Technology, Chennai, India
[6]Department of Science and Humanities (General Engineering-EEE), R.M.K. College of Engineering and Technology, Puduvoyal, India

Abstract

Coronary artery disease (CAD) is a leading cause of mortality worldwide, necessitating early, accurate, and non-invasive prediction methods. While machine learning offers a promising alternative to conventional invasive diagnostics, challenges such as high-dimensional data, class imbalance, and redundant features remain. This study proposes HESML-CAD, a hybrid ensemble stacking and meta-learning framework enhanced with Bald Eagle Search Optimization (BESO) and explainable AI for reliable and interpretable CAD prediction. The model was evaluated on two datasets: Framingham (4200 samples, 15 features) for 10-year risk prediction and Z-Alizadeh Sani (304 samples, 55 features) for current diagnosis. Preprocessing included imputation, normalization, encoding, and ADASYN-based balancing. Feature selection combined Chi-square, mutual information, autoencoder-based reduction, and BESO optimization. Five base classifiers (KNN, SVM-RBF, Random Forest, XGBoost, and CatBoost) were stacked using Logistic Regression, with Platt scaling for calibration and SHAP for explainability.On the Framingham dataset, HESML-CAD achieved an accuracy of 0.93, F1-score of 0.93, and AUC of 0.96, outperforming the best individual model (Random Forest: accuracy 0.896, F1-score 0.90) by approximately 3–4%, with a low Brier score of 0.0540. On the Z-Alizadeh Sani dataset, it achieved an accuracy of 0.94, F1-score of 0.935, and AUC of 0.98, surpassing strong baselines (accuracy $\approx$ 0.93), while maintaining well-calibrated probability estimates (Brier score 0.0547). SHAP identified clinically relevant predictors such as age, systolic blood pressure, and chest pain. Overall, HESML-CAD provides accurate, calibrated, and interpretable CAD predictions, demonstrating potential for supporting non-invasive clinical screening.

Keywords Coronary artery disease, Ensemble learning, Feature selection, Explainable AI, Risk prediction

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1 Introduction

Coronary artery disease (CAD) remains a major contributor to global morbidity and mortality, significantly affecting the overall prevalence of cardiovascular diseases [43]. It develops gradually due to the buildup of atherosclerotic plaque, which causes narrowing or blockage of the coronary arteries and reduces oxygen-rich blood flow to the heart muscle [26]. This condition can lead to serious complications, including myocardial infarction, heart failure, and sudden cardiac death. Early and accurate detection of CAD is essential for slowing its progression, improving patient outcomes, and reducing healthcare costs [44]. Although diagnostic techniques such as electrocardiography (ECG), echocardiography, stress testing, and coronary angiography are available, they are often invasive and expensive [20]. Consequently, machine learning (ML) methods have gained increasing attention as non-invasive, affordable, data-driven tools for assessing CAD risk, leveraging routinely collected clinical and demographic data to support timely decision-making [19].

ML has become a vital technology in healthcare, enabling the extraction of valuable insights from large and heterogeneous clinical datasets [30]. For CAD prediction, ML models utilize diverse patient information, including demographics, lifestyle factors, laboratory results, and clinical symptoms [1]. Several algorithms, including KNN, SVM, Random Forest, and gradient boosting methods, have been successfully applied for CAD classification [33]. In addition, feature selection techniques such as LASSO, Random Forest-based selection, and Boruta have been used to identify clinically relevant predictors, such as the TyG-index, as reliable markers of CAD risk [31]. However, these models often face challenges related to high dimensionality, redundant features, and multicollinearity, which can lead to overfitting and reduced interpretability in clinical settings [36].

Data preprocessing plays a crucial role in preparing clinical datasets for ML applications. Common practices include handling missing values using median imputation, scaling numerical features, encoding categorical variables, and addressing class imbalance using techniques such as Adaptive Synthetic Sampling (ADASYN) [24, 37]. Recent studies have increasingly adopted hybrid feature selection approaches that combine statistical filter methods (e.g., Chi-square and Mutual Information) with deep learning-based techniques such as autoencoders to capture complex, non-linear relationships [5, 10]. Furthermore, meta-heuristic optimization algorithms, such as Bald Eagle Search Optimization (BESO), have been introduced to efficiently explore high-dimensional feature spaces and identify informative feature subsets [11, 12]. These approaches help improve generalization, reduce redundancy, and enhance predictive performance in CAD risk modeling [6, 25].

This study presents HESML-CAD as an integrative machine learning framework for coronary artery disease prediction, combining feature selection, ensemble learning, and explainability within a unified pipeline. Rather than introducing a new standalone algorithm, the contribution lies in the systematic design and clinical alignment of these components to address key challenges such as high dimensionality, class imbalance, and interpretability. The framework incorporates hybrid feature selection using statistical methods, autoencoder-based reduction, and BESO optimization, followed by a stacked ensemble model with a meta-learner and probability estimation. SHAP-based explainability is integrated to provide clinically meaningful insights into model predictions.

The main objectives are: (a) to investigate the role of BESO in optimizing feature subsets and improving predictive performance, (b) to evaluate and compare machine learning models on two clinically distinct CAD datasets, and (c) to identify a robust and interpretable approach for non-invasive CAD prediction. The main contributions of this study are: (i) the development of an integrated HESML-CAD pipeline combining preprocessing, feature selection, ensemble learning, calibration, and explainability; (ii) the application of a hybrid feature selection strategy that balances dimensionality reduction and predictive relevance; (iii) the design of a stacking-based ensemble framework for improved generalization across heterogeneous datasets; and (iv) the incorporation of explainability and calibration to support clinically meaningful interpretation of predictions.

The paper is organized as follows: Sect. 2 reviews related works, Sect. 3 describes the methodology, Sect. 4 presents the experimental analysis and findings, Sect. 5 discusses the results, Sect. 6 outlines the strengths and limitations, and Sect. 7 concludes the study.

2 Related works

In recent years, ML has gained considerable attention for the non-invasive prediction of CAD. Several comparative and review studies have reported that models such as SVM, RF and boosting methods demonstrate promising discriminative performance in cardiovascular disease prediction tasks [22]. However, these studies often remain limited in their ability to address challenges such as high-dimensionality, class imbalance, model interpretability, and probability calibration.

Specifically, various approaches have been proposed for CAD classification. Akella et al. [4] applied several ML classifiers, including SVM, random forest, and logistic regression, to clinical datasets for CAD prediction and demonstrated that ML-based models can outperform traditional statistical approaches. Similarly, Mohammedqasim et al. [32] introduced the HEVO framework, an ensemble voting model applied to the Z-Alizadeh Sani dataset, which improved classification performance over single classifiers. Kolukisa et al. [21] further advanced this direction by combining ensemble feature selection with classification, highlighting that hybrid feature selection strategies can enhance predictive accuracy. More recently, Olawade et al. [35] incorporated BESO into CAD prediction, demonstrating that it significantly improved performance across multiple models.

Recent advances in deep learning have further expanded the scope of computational modeling in biomedical prediction tasks. Several studies have proposed sophisticated architectures integrating sequence encoding, image-based representations, and capsule networks for improved predictive performance. For instance, the iSucc-SnCNs model employs transformer-based embeddings combined with a self-normalized capsule network for protein modification prediction, achieving strong generalization across datasets. Similarly, TargetAVP-DeepCaps integrates multiperspective feature encoding with capsule networks to enhance antiviral peptide prediction accuracy, while DeepAIPs-SFLA combines deep convolutional neural networks with evolutionary feature representations and metaheuristic optimization for anti-inflammatory peptide prediction. In addition, capsule network-based approaches such as pACPCapsNet demonstrate the effectiveness of sequence-to-image transformations in capturing complex biological patterns. These models highlight the growing trend of combining deep feature extraction,

metaheuristic optimization, and advanced neural architectures to improve predictive performance in biomedical domains [48–51].

More broadly, recent studies have emphasized that model selection and evaluation in complex prediction tasks must consider dataset characteristics and metric sensitivity, as no single model or evaluation strategy consistently performs best across all domains [52]. These findings reinforce the need for integrative and adaptable frameworks in healthcare applications.

Beyond CAD diagnosis, several works have addressed cardiovascular risk prediction more broadly. For example, Cicek et al. [8] conducted a systematic review of ML applications on SPECT data for CAD prognosis, observing that integrating imaging and demographic features often yields stronger predictive models, though many lacked calibration and interpretability. Cheng et al. [7] applied gradient boosting to a Taiwanese biobank cohort for CAD outcome prediction, identifying age and biochemical markers as key predictors; however, their work did not thoroughly explore ensemble stacking or hybrid feature optimization.

Despite these advances, several critical gaps persist in the literature. Many studies rely on conventional feature selection techniques such as LASSO, univariate filters, or recursive feature elimination (RFE), which often fail to capture nonlinear dependencies and may introduce redundancy [17]. Ensemble learning is sometimes applied, but typically in the form of simple majority voting, while more sophisticated stacking-based meta-learning approaches remain underexplored in CAD prediction [23]. Furthermore, although most studies report discrimination metrics such as AUC and accuracy, few address probability calibration through measures like Brier score or Platt scaling, thereby limiting clinical trust in the predicted risk estimates [38, 39]. Model interpretability also remains a significant challenge: while some works report feature importance scores, robust explainable AI approaches such as SHAP, which provide both global and local interpretability, are rarely employed [27]. Finally, external and cross-dataset validation are uncommon, as most models are evaluated on a single dataset, which limits their ability to demonstrate consistent performance across different data sources [9].

The proposed HESML-CAD framework addresses these shortcomings systematically. To overcome the limitations of conventional feature selection, we integrate filter-based methods (Chi-square, mutual information), deep feature reduction through autoencoders, and metaheuristic optimization using the BESO algorithm, which has recently demonstrated strong performance in biomedical data analysis [3]. By adopting a stacked ensemble architecture that combines multiple heterogeneous base classifiers with a logistic regression meta-learner, our framework surpasses the simplicity of majority-voting approaches and exploits the complementary strengths of diverse models [45]. We further enhance clinical reliability by incorporating probability calibration with Platt scaling, ensuring that predicted probabilities are well-aligned with observed outcomes—an aspect often overlooked in cardiovascular ML studies [28]. To enhance transparency, we utilize SHAP-based analysis at both the global and local levels, allowing clinicians to understand the relative contributions of features such as age, systolic blood pressure, and diabetes, thereby fostering confidence in model predictions [29]. Collectively, these innovations ensure that HESML-CAD provides accurate, calibrated, and interpretable CAD predictions while maintaining generalizability across diverse clinical populations.

3 Methodology

This study presents an integrated machine learning framework, termed HESML-CAD, for predicting coronary artery disease. The framework consists of multiple stages, including dataset collection, data preprocessing, hybrid feature selection using filter methods, autoencoder-based dimensionality reduction, and Bald Eagle Search Optimization (BESO), followed by ensemble model training and evaluation. The modeling process incorporates five base classifiers and a logistic regression meta-learner, along with probability estimation and explainable AI techniques to enhance interpretability. The overall workflow is illustrated in Fig. 1.

3.1 Data collection

This study employed two publicly accessible datasets: the Framingham Heart Study dataset and the Z-Alizadeh Sani dataset. These datasets were selected to represent both population-level long-term CAD risk prediction [40] and clinically detailed current CAD diagnosis, enabling a comprehensive evaluation of the HESML-CAD framework.

The Framingham dataset is derived from a well-established longitudinal cohort study investigating cardiovascular risk factors. It consists of 4,200 instances with 15 features, including demographic variables (age, sex), lifestyle factors (smoking, physical activity), clinical measures (blood pressure, cholesterol), and medical history (diabetes, stroke). The target variable represents the 10-year risk of developing CAD based on longitudinal follow-up, supporting long-term predictive modeling.

In contrast, the Z-Alizadeh Sani dataset was collected from patients at the Shaheed Rajaei Cardiovascular, Medical, and Research Centre in Tehran. It contains 304 records with 55 features encompassing demographic data, symptom profiles (e.g., typical or atypical chest pain), physical examination findings, ECG parameters, laboratory markers, and echocardiographic measurements. The target variable indicates the presence or absence of CAD, defined as ≥ 50% stenosis confirmed through angiography, which serves as the clinical reference standard.

Together, these datasets differ in sample size, feature complexity, and prediction objectives, providing a comprehensive basis for evaluating the proposed framework. Table 1 summarizes the characteristics of the datasets used in this study.

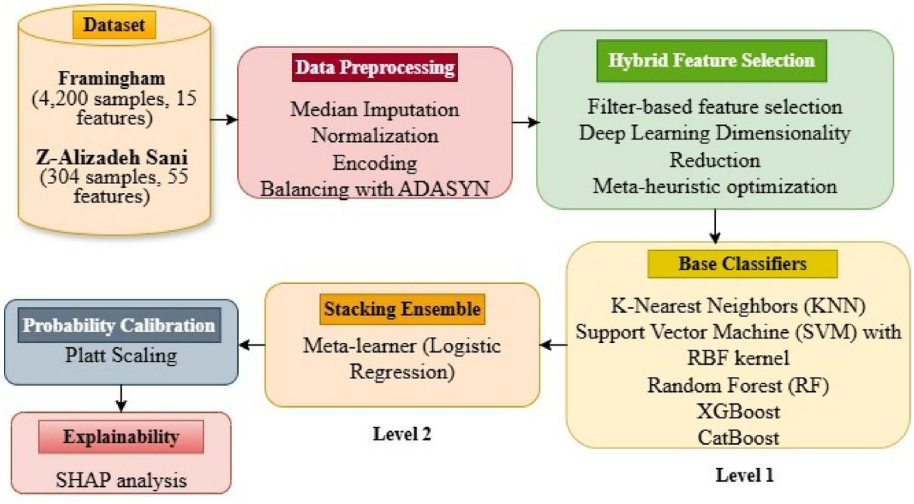


Fig. 1 Architecture of the HESML-CAD Framework

3.2 Clinical background of coronary artery disease and diagnostic criteria

Coronary artery disease (CAD) is characterized by the progressive narrowing or obstruction of coronary arteries due to atherosclerotic plaque accumulation, leading to reduced myocardial blood flow and an increased risk of ischemic events such as angina pectoris and myocardial infarction. Clinically, CAD diagnosis and risk assessment rely on a combination of demographic factors, lifestyle habits, physiological measurements, biochemical markers, imaging findings, and, in some cases, invasive investigations, depending on the clinical context.

In the Framingham dataset, CAD is defined as a long-term risk outcome, representing the 10-year probability of developing coronary events, including coronary insufficiency, myocardial infarction, or angina pectoris. The reference standard in this cohort is longitudinal clinical follow-up, where outcomes are determined over multiple examination cycles. The data modality is primarily population-level epidemiological data, including demographic attributes (age, sex), lifestyle factors (smoking), and routine clinical measurements (blood pressure, cholesterol levels, diabetes status). Thus, Framingham represents a prospective risk prediction task rather than direct disease diagnosis.

In contrast, the Z-Alizadeh Sani dataset focuses on current CAD diagnosis. The reference standard is coronary angiography, with CAD defined as ≥ 50% stenosis in at least one major coronary artery. This dataset integrates multiple data modalities, including demographic information, symptom characteristics (e.g., typical chest pain), electrocardiographic findings (ST-segment changes, T-wave inversion), laboratory measurements (fasting blood sugar, triglycerides, creatinine), and echocardiographic parameters (ejection fraction, regional wall motion abnormality). Therefore, it represents a cross-sectional diagnostic classification task closely aligned with real-world cardiology practice.

The distinction between prospective risk estimation (Framingham) and angiography-confirmed diagnosis (Z-Alizadeh Sani) is critical for interpreting model behavior, feature relevance, and performance outcomes. It also explains why different machine learning models exhibit varying strengths across the two datasets and supports the use of an ensemble-based framework capable of adapting to heterogeneous clinical context.

3.3 Data preprocessing

A structured preprocessing pipeline was implemented in the HESML-CAD framework to ensure data quality, consistency, and suitability for machine learning modeling. To prevent information leakage, all preprocessing steps were performed exclusively within the training data during cross-validation, and the learned transformations were subsequently applied to validation and test sets. The main steps are described below.

Table 1 Summary of datasets

Dataset	Source	Samples	Features	Feature types	Prediction task	Target variable
Framingham heart study	Longitudinal cohort (USA)	4200	15	Demographic, lifestyle, clinical, medical history	10-year CAD risk prediction (long-term risk)	10-year CAD outcome (yes/no)
Z-Alizadeh Sani	Shaheed Rajaei Cardiovascular Center, Iran	304	55	Demographic, symptoms, physical exam, ECG, labs, echo	CAD diagnosis (cross-sectional classification)	CAD presence (angiography ≥ 50%)

3.3.1 Handling missing values

We applied median imputation to address missing values. For example, if a patient's systolic blood pressure value was missing in the Framingham dataset, the median blood pressure across the training samples was used as a substitute. This approach reduces the influence of outliers while preserving the central tendency of each feature distribution [41], thereby maintaining data integrity.

3.3.2 Feature normalization

To standardize numerical features, we used the StandardScaler function, which transforms data to have a mean of zero and a standard deviation of one. For instance, cholesterol levels measured in mg/dL were normalized so that extreme values did not disproportionately influence model training. Standardization ensured uniform scaling across features, which is particularly important for distance-based models such as KNN and for improving convergence in gradient-based algorithms.

3.3.3 Categorical encoding

Categorical variables were encoded using Scikit-learn'sLabelEncoder. For example, the variable sex (male/female) was converted into integer values (0/1). This transformation enabled machine learning algorithms to process categorical data efficiently without introducing additional dimensionality.

3.3.4 Class imbalance handling

To address class imbalance between CAD-positive and CAD-negative cases, we employed the Adaptive Synthetic Sampling (ADASYN) technique. For example, in the Z-Alizadeh Sani dataset, where CAD-negative cases outnumber CAD-positive cases, ADASYN generated synthetic samples of the minority class by focusing on difficult-to-classify instances.

To ensure unbiased evaluation, oversampling was applied only to the training data within each fold, preventing leakage of synthetic information into validation or test sets.

Table 2 illustrates examples of preprocessing transformations applied in the HESML-CAD framework. These steps ensured that the input data was cleaned, standardized, and balanced, providing a reliable foundation for subsequent feature selection and model training.

3.4 Feature selection

To enhance model performance, reduce overfitting, and improve computational efficiency, the HESML-CAD pipeline employed a hybrid feature selection strategy. This strategy integrated three complementary approaches: filter-based selection, deep learning-based dimensionality reduction, and meta-heuristic optimization. To prevent

Table 2 Example of preprocessing transformations applied in the HESML-CAD framework

Stage	Feature (Example)	Before preprocessing	After preprocessing
Handling missing values	Systolic BP (mmHg)	Missing value (NaN)	Imputed with dataset median (e.g., 128 mmHg)
Feature normalization	Cholesterol (mg/dL)	280	1.34 (standardized z-score)
Categorical encoding	Sex	Male / Female	1 / 0
Class imbalance handling	CAD Diagnosis (Z-Alizadeh)	CAD-positive: 85, CAD-negative: 219 (raw)	CAD-positive: 219, CAD-negative: 219 (after ADASYN)

information leakage and ensure unbiased performance evaluation, all feature selection steps were performed strictly within each training fold during cross-validation. These components operate in a sequential pipeline rather than a parallel or voting-based scheme. Specifically, filter-based methods first identify relevant features, the autoencoder then learns compact nonlinear representations from the reduced feature set, and finally, the BESO algorithm selects the optimal subset of features. This staged process enables progressive refinement of the feature space, improving both representation quality and model generalization.

3.4.1 Filter-based feature selection

We first applied statistical filter techniques to assess the relevance of features with respect to the target variable. Specifically, the Chi-square test and Mutual Information were used to rank features based on their dependency on the outcome class. These computations were performed exclusively on the training portion of each fold, and the resulting feature rankings were then applied to the corresponding validation fold. These rankings provided a preliminary guide for selecting informative features.

3.4.2 Deep learning-based dimensionality reduction

Next, we utilized an autoencoder, an unsupervised neural network, to learn a compressed representation of the input data by minimizing reconstruction error. The encoded layer captured salient nonlinear interactions [46], effectively reducing dimensionality while preserving essential information for CAD prediction.

The autoencoder architecture consisted of a symmetric feedforward network with an input layer matching the number of features, followed by two hidden encoding layers (with progressively reduced dimensions), a bottleneck layer representing the compressed feature space, and corresponding decoding layers to reconstruct the input. Rectified Linear Unit (ReLU) activation functions were used in the hidden layers, while a linear activation was applied in the output layer. The network was trained using the Adam optimizer with mean squared error (MSE) as the loss function.

The autoencoder was trained only on the training data within each fold, and the learned encoding was subsequently applied to the validation data, ensuring that no information from the validation set influenced the representation learning process.

3.4.3 Meta-heuristic optimization with BESO

Finally, to refine the feature subset, we applied the BESO algorithm [13]. Inspired by the foraging dynamics of bald eagles, BESO efficiently balances exploration and exploitation to identify high-performing feature combinations. BESO optimization was conducted using only the training subset of each fold, and the selected feature subset was then evaluated on the corresponding validation fold. Unlike traditional methods such as LASSO regression, BESO avoids local optima and enhances model generalizability. Algorithm 1 illustrates the feature subsets selected using BESO within the HESML-CAD framework.

Input:Features X, Target y for the dataset
*N*Population size
*MaxIter*Maximum iterations
*D*Search space dimensions
Fitness function
Parameters: α, β, c_1, c_2 (control exploration and exploitation)
Output: Feature subset (optimal)
begin
1.Establish Population:
Generate *N* random binary vectors of length *D*
1 = feature selected
0 = feature not selected
2. EvaluateFitness:
For each outcome:
Select features based on binary vector
Train base model
3.Compute fitness using:
$Fitness = \alpha \cdot (1 - F1\text{-}score) + \beta \frac{|S|}{|T|}$
where |S| = number of selected features, |T| = total features
Optimization by iteration (Repeat for *iteration till MaxIter):*
4.Select Exploration:
Compute the mean position of all solutions
Update each solution with a movement influenced by the mean and random perturbations.
5.Search Space (Global Search):
Identify the current best solution (highest fitness)
Guide the movement of other solutions toward this best solution with added randomness
6.Swoop Space (Exploitation):
Simulate the bald eagle's swoop toward the target
Update each solution using a combination of the best solution, previous position, and random direction
7.Re-evaluate Fitness:
Recompute fitness for all solutions
Update global best if improved:
Output the feature subset
End

Algorithm 1 HESML-CAD feature selection using BESO

The integration of these three stages enabled the HESML-CAD framework to identify compact and relevant feature subsets, contributing to improved representation quality and model performance. However, the overall predictive gains are attributed to the combined effect of feature selection, ensemble learning, and calibration within the integrated framework.

3.5 Machine learning algorithms for HESML-CAD

The predictive core of the HESML-CAD framework is built on a stacked ensemble architecture that integrates multiple heterogeneous base classifiers (Level 1) with a logistic regression meta-learner (Level 2). This design was deliberately chosen to leverage the complementary strengths of different learning paradigms, ensuring robustness across datasets with varying feature dimensions and class imbalance.

3.5.1 Base classifiers (Level 1)

We employed five diverse classifiers, each offering distinct inductive biases that address specific challenges in CAD prediction. KNN [15] provides strong local decision boundaries and is sensitive to distance metrics, making it effective in capturing patient similarity patterns. SVM-RBF [14] excels in handling non-linearly separable cases, particularly relevant for complex biomedical relationships. RF [18] introduces variance reduction

through bagging and random feature selection, improving robustness in noisy or high-dimensional clinical data. XGB [34], a gradient boosting technique, optimizes predictive accuracy by sequentially correcting errors of weak learners, and has proven effective on structured medical datasets. CB [16] enhances boosting by handling categorical variables natively and incorporating ordered boosting, thereby reducing overfitting and providing strong generalization. The inclusion of these models ensured diversity in decision-making strategies, ranging from distance-based to kernel-based and tree-based learners.

3.5.2 Meta-learner (Level 2) and model calibration

The outputs (predicted probabilities) of the base classifiers were used as input features for a logistic regression (LR) meta-learner. Unlike simple majority voting, this stacking strategy learns optimal combinations of base model outputs, enabling it to leverage complementary decision boundaries for improved predictive performance.

To further support probability interpretation, Platt scaling was applied to the meta-learner outputs. While the model was already reasonably well-calibrated prior to this step, calibration was retained to improve probability alignment in specific regions of the prediction space, which is relevant for clinical risk interpretation.

This ensemble strategy was designed not only to improve predictive performance but also to maintain model stability and interpretability. By combining diverse classifiers and incorporating explainable AI techniques, the HESML-CAD framework provides a balanced trade-off between accuracy and clinical applicability.

3.6 Performance evaluation

To ensure a robust and clinically meaningful assessment of the HESML-CAD framework, we implemented a structured evaluation protocol integrating data partitioning, hyperparameter tuning, holdout testing, and probability calibration. This multi-stage strategy was designed to assess predictive accuracy, discrimination ability, calibration quality, and generalization performance while minimizing the risk of information leakage.

3.6.1 Train-test split

Each dataset was randomly partitioned into training (80%) and testing (20%) subsets using stratified sampling to preserve class distribution, particularly for the imbalanced Z-Alizadeh Sani dataset. The independent test set was strictly held out and used only for final evaluation.

All preprocessing steps—including missing value imputation, feature scaling, categorical encoding, class balancing using ADASYN, and hybrid feature selection—were performed exclusively within the training data. No information from the test set was used during preprocessing, feature selection, or model training. The generalization ability of the HESML-CAD framework was assessed using this independent test set.

3.6.2 Empirical hyperparameter optimization

We tuned the hyperparameters of each base classifier using grid search combined with fivefold cross-validation on the training set. Within each fold, all preprocessing and

feature selection steps were re-fitted exclusively on the training portion, ensuring strict isolation between training and validation data and preventing information leakage.

For the stacking framework, a leakage-free training strategy was employed. Base learners were trained using cross-validation, and their out-of-fold predictions were used to construct the meta-learner training data. This ensures that the meta-learner does not receive predictions from models trained on the same samples.

From a broader methodological perspective, this repeated sampling strategy aligns with distribution-based evaluation principles discussed in the A-Test framework [47], although it is implemented here as standard cross-validation rather than a standalone A-Test procedure. Optimal configurations were selected based on F1-score and AUC to ensure a balanced trade-off between sensitivity and specificity.

3.6.3 Holdout evaluation

The optimized models were evaluated on the independent holdout test set using a comprehensive suite of metrics [2]:

- Accuracy: overall correctness of predictions
- Precision: proportion of predicted positives that are true positives
- Recall (Sensitivity): proportion of actual CAD cases correctly identified
- F1-score: harmonic mean of precision and recall
- AUC-ROC: discriminative ability across classification thresholds
- Brier Score: calibration quality based on probability error

To quantify statistical reliability, 95% confidence intervals were computed using bootstrap resampling (1,000 iterations) on the holdout test predictions, providing robust estimates of performance variability.

3.6.4 Probability calibration

To support probability interpretation, Platt scaling was applied to calibrate the meta-learner's outputs. While calibration effects were modest, this step provides improved probability alignment in specific regions of the prediction space, which is relevant for clinical risk interpretation.

These validation strategies ensure that the HESML-CAD pipeline is rigorously evaluated, minimizes information leakage, and provides reliable performance estimates across heterogeneous clinical datasets.

3.7 Evaluation metrics

To thoroughly evaluate the performance of the HESML-CAD framework on both datasets, we employed a comprehensive set of standard classification metrics, each of which is formally defined below, Eqs. (1) to (8) respectively.

Accuracy represents the overall proportion of correct predictions, reflecting general predictive effectiveness:

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)} \tag{1}$$

where TP = true positives, TN = true negatives, FP = false positives, and FN = false negatives.

Precision quantifies the proportion of true positives among all positive predictions, indicating the model's ability to minimize false alarms:

$$Precision = \frac{TP}{(TP + FP)} \tag{2}$$

Recallmeasures the proportion of actual CAD cases correctly identified, capturing the model's effectiveness in detecting positive instances:

$$Recall = \frac{TP}{(TP + FN)} \tag{3}$$

F1-score is the harmonic mean of precision and recall, balancing both measures and offering robustness for imbalanced datasets:

$$F1 - Score = 2 \times \frac{Precision * Recall}{Precision + Recall} \tag{4}$$

True Positive rate is calculated using

$$TruePosiiveRate(TPR) = \frac{TP}{(TP + FN)} \tag{5}$$

False Positive rate is calculated using

$$FalsePosiiveRate(FPR) = \frac{FP}{(FP + TN)} \tag{6}$$

Area Under the Receiver Operating Characteristic Curve (AUC-ROC) evaluates the discriminative ability of the model across various classification thresholds:

$$AUC - ROC = \int_0^1 TPR(FPR)d(FPR) \tag{7}$$

Brier Score measures the mean squared difference between predicted probabilities and actual outcomes, providing insight into calibration quality:

$$BrierScore = \frac{1}{N}\sum_{i=1}^{N}(p_i - y_i)^2 \tag{8}$$

where p_i is the predicted probability for instance i, $y_i \in \{0,1\}$ is the true label, and N is the total number of samples.

All evaluation metrics were computed on the independent test set and are reported with 95% confidence intervals, ensuring statistical rigor and reliability of performance estimates.

3.8 Baseline comparisons

To rigorously evaluate the effectiveness of the BESO algorithm in feature selection, we established a comprehensive set of baseline models. As a primary benchmark, classifiers were trained using all available features from each dataset without any reduction, allowing us to assess the raw performance of models on high-dimensional input data. This

provided a reference point to evaluate whether feature selection genuinely improved model efficiency and accuracy.

We then compared BESO against several widely used conventional feature selection techniques. Filter-based methods, including the Chi-squared test and Mutual Information, ranked features based on their statistical association with the target variable, capturing simple dependencies but often ignoring feature interactions. As a wrapper approach, RFEwas employed to select subsets of features based on model performance iteratively, providing a performance-driven selection but at a high computational cost. Finally, the embedded method of Least Absolute Shrinkage and Selection Operator (LASSO) regularization was implemented, which inherently selects features by penalizing less informative variables during model training.

By contrasting BESO with these methods, we aimed to evaluate its effectiveness within the overall HESML-CAD framework. However, it is important to note that the observed performance differences reflect the combined influence of feature selection, ensemble learning, and calibration, rather than the isolated effect of BESO alone. This comparative setup provides a practical benchmark but does not constitute a fully controlled ablation of individual components.

3.9 Comparison with clinical risk scores

To ensure clinical relevance and contextualize the performance of the HESML-CAD framework, we conducted comparative evaluations against three widely recognized cardiovascular risk assessment tools. For the Framingham dataset, we implemented the updated Framingham Risk Score (FRS), which estimates 10-year cardiovascular risk using demographic and clinical variables. The Systematic Coronary Risk Evaluation (SCORE) model, widely used in Europe, was also included to predict the 10-year risk of fatal cardiovascular disease, offering a region-specific comparator. Finally, the ASCVD Risk Estimator Plus, developed by the American College of Cardiology and the American Heart Association, was employed to predict both 10-year and lifetime atherosclerotic cardiovascular disease risk.

These risk calculators were evaluated using the same test partitions and performance metrics—Accuracy, Precision, Recall, F1-score, AUC, and Brier Score—as applied to the machine learning models. However, it is important to note that these tools were originally developed for different populations, endpoints, and clinical use cases. In particular, the Framingham Risk Score and ASCVD estimator are designed for long-term risk prediction, while the Z-Alizadeh Sani dataset represents a cross-sectional diagnostic setting. Therefore, this comparison should be interpreted as an exploratory benchmark rather than a direct clinical equivalence. Under these conditions, HESML-CAD demonstrated improved predictive performance in terms of discrimination and classification metrics, although the comparison does not imply superiority within the original intended clinical use of these risk scores.

3.9.1 Target variable definition

To ensure clinical interpretability and methodological clarity, we explicitly defined the prediction objectives for each dataset used in this study. In the Framingham dataset, the target variable corresponds to the 10-year likelihood of developing CAD, including coronary insufficiency, myocardial infarction, and angina pectoris. This variable is derived

from longitudinal follow-up protocols [42], framing the task as a prospective risk estimation problem where the goal is to identify individuals at high risk before the onset of disease.

In contrast, the Z-Alizadeh Sani dataset defines CAD as the current presence of ≥ 50% stenosis, verified through angiographic imaging, which serves as the clinical gold standard for diagnosis. This dataset therefore supports a cross-sectional disease classification task aimed at diagnosing CAD in real-time.

Distinguishing between these two prediction objectives—prospective risk prediction (Framingham) versus real-time diagnosis (Z-Alizadeh Sani)—is critical for interpreting model outputs, selecting appropriate evaluation metrics, and ensuring clinical relevance. Accordingly, all reported results, including discrimination and calibration metrics, were contextualized within the specific prediction goal of each dataset.

4 Experimental analysis and findings

In this study, the Framingham and Z-Alizadeh Sani datasets were employed to evaluate the proposed HESML-CAD framework for CAD prediction, incorporating hybrid feature selection with BESO. The experimental results demonstrate that the overall framework achieves strong predictive performance across both datasets. While BESO contributes to identifying compact feature subsets, the observed performance improvements should be interpreted as the result of the integrated HESML-CAD pipeline rather than the isolated effect of any single component.

We conducted a detailed evaluation by comparing individual classifiers—namely KNN, SVM-RBF, Random Forest, XGBoost, and CatBoost—alongside a stacked ensemble model using logistic regression as the meta-learner. To ensure reproducibility and consistency, each model was tuned using grid search with fivefold cross-validation, and optimal hyperparameters were selected based on F1-score and AUC. For KNN, $k = 7$ with Euclidean distance balanced sensitivity and specificity. For SVM-RBF, $C = 10$ and $\gamma = 0.01$ provided a suitable trade-off between margin maximization and generalization. For Random Forest, n_estimators = 200 and max_depth = 12 ensured stable performance. For XGBoost, learning_rate = 0.05, max_depth = 6, and n_estimators = 300 achieved reliable results. For CatBoost, depth = 8, iterations = 500, and learning_rate = 0.1 provided consistent performance. The logistic regression meta-learner employed L2 regularization ($C = 1.0$) to effectively combine base model predictions.

Performance was assessed using accuracy, precision, recall, F1-score, AUC-ROC, and Brier score, providing a comprehensive evaluation of classification performance, discrimination, and calibration. The results indicate that the stacked ensemble generally achieves improved and more stable performance compared to individual classifiers, highlighting the benefit of combining diverse learning models. Overall, the integration of feature selection, ensemble learning, and calibration contributes to improved predictive consistency and reduced redundancy within the proposed framework.

4.1 Performance comparison of ML models on CAD prediction for framingham dataset

Table 3 summarizes the experimental results for the Framingham dataset. Using the proposed hybrid feature selection pipeline, 15 optimal features were selected from the original feature set, including clinically relevant predictors such as age, systolic blood

Table 3 Experimental results of ML models for the Framingham dataset

Dataset	Model	Accuracy	Precision	Recall	F1 Score	AUC	Brier Score
Framingham	KNN	0.789	0.715	0.975	0.825	0.902	–
	SVM-RBF	0.695	0.682	0.755	0.717	0.772	–
	Random Forest	0.896	0.883	0.916	0.9	0.964	–
	XGBoost	0.893	0.914	0.873	0.893	0.953	–
	CatBoost	0.891	0.926	0.856	0.89	0.949	–
	HESML-CAD (Meta-learner)	0.93	0.93	0.93	0.93	–	0.054

Table 4 Experimental results of ML models for the Z-Alizadeh Sani dataset

Dataset	Model	Accuracy	Precision	Recall	F1 Score	AUC	Brier score
Z-Alizadeh Sani	KNN	0.82	0.727	0.966	0.83	0.948	–
	SVM-RBF	0.938	0.891	0.983	0.934	0.96	–
	Random Forest	0.93	0.93	0.914	0.922	0.977	–
	XGBoost	0.93	0.915	0.931	0.923	0.968	–
	CatBoost	0.93	0.889	0.966	0.926	0.977	–
	HESML-CAD (Meta-learner)	0.94	0.9	0.97	0.935	–	0.0547

pressure, diabetes status, total cholesterol, and smoking history. These features were used across all model configurations to ensure consistency in evaluation.

The results indicate that the HESML-CAD stacked ensemble achieved the strongest overall performance, with an accuracy and F1-score of 0.93, an AUC of 0.96, and a Brier score of 0.054, demonstrating strong discriminative ability along with reliable probability estimates. Among individual classifiers, Random Forest (RF) emerged as the best performer, achieving an F1-score of 0.90 and an AUC of 0.964. Its robustness in handling non-linearity, capturing complex feature interactions, and resisting overfitting makes it particularly effective for CAD prediction in population-based datasets such as Framingham.

In comparison, boosting models (XGBoost and CatBoost) also demonstrated strong and consistent results, with AUC values above 0.94, reflecting their ability to balance bias and variance through sequential learning. However, their precision–recall trade-offs were slightly less balanced than RF. KNN and SVM-RBF showed comparatively lower performance, with F1-scores of 0.825 and 0.717, respectively. KNN achieved high recall (0.975) but lower precision, indicating a higher rate of false positives. SVM-RBF improved upon linear models but showed sensitivity to parameter tuning and variability in clinical data.

The findings suggest that while individual ensemble models such as RF, XGBoost, and CatBoost provide strong predictive performance, the stacked ensemble (HESML-CAD) achieves improved and more stable results by combining complementary model strengths. The relatively low Brier score further indicates that the model produces reasonably well-calibrated probability estimates, which are important for clinical interpretation.

4.2 Performance comparison of ML models on CAD prediction for Z-Alizadeh Sani dataset

Table 4 presents the performance of machine learning models on the Z-Alizadeh Sani dataset. Using the proposed hybrid feature selection pipeline, a compact subset of 10 features was selected from the original 55, including clinically informative predictors such as typical chest pain, EF-TTE, age, region RWMA, triglycerides, fasting blood

sugar, T-inversion, hypertension, ESR, and diabetes mellitus. These features were consistently used across all models to ensure fair comparison.

The HESML-CAD stacked ensemble achieved strong overall performance, with an accuracy of 0.94, F1-score of 0.935, AUC of 0.98, and a Brier score of 0.0547. These results indicate high discriminative ability along with reasonably reliable probability estimates for CAD diagnosis.

Among the base learners, SVM with an RBF kernel exhibited the highest recall (0.983), making it effective in minimizing false negatives, which is particularly important in diagnostic settings. However, its slightly lower precision (0.891) compared to Random Forest and boosting models reflects a trade-off with false positives. Random Forest, CatBoost, and XGBoost delivered highly competitive performance, with F1-scores above 0.92 and AUC values exceeding 0.96, demonstrating their robustness in handling high-dimensional clinical data. CatBoost showed strong recall (0.966), highlighting its effectiveness in capturing diagnostic patterns.

In contrast, KNN achieved high recall (0.966) but relatively lower precision (0.727), indicating sensitivity to false positives. Its dependence on local neighborhood structure makes it less stable in smaller and complex datasets.

The results suggest that the hybrid feature selection process contributes to reducing dimensionality while retaining clinically meaningful predictors. The stacked ensemble (HESML-CAD) further benefits from combining diverse model outputs, resulting in improved and more stable performance across metrics. The relatively low Brier score indicates that the model provides reasonably calibrated probability estimates, which are important for clinical interpretation.

4.3 Brier score and calibration

In clinical risk prediction tasks such as CAD, it is essential not only to classify patients correctly but also to provide well-calibrated probability estimates that accurately reflect the likelihood of disease presence. To address this, the proposed HESML-CAD framework incorporates Brier score evaluation and Platt scaling for probability calibration. Table 5 shows the calibration results for both datasets under study.

In our experiments, the HESML-CAD model achieved low Brier scores of 0.0540 (Framingham) and 0.0547 (Z-Alizadeh Sani), indicating reasonably well-calibrated probability estimates and supporting the framework's applicability for non-invasive CAD risk screening.

To assess and potentially refine calibration, Platt scaling was applied to the output of the meta-learner. This method fits a logistic regression model to the prediction scores, transforming them into calibrated probabilities that are clinically interpretable.

To evaluate the reliability of predicted probabilities generated by the HESML-CAD framework, calibration curves were plotted for both the Framingham and Z-Alizadeh Sani datasets. These curves illustrate how closely the predicted probabilities align with the actual outcomes, which is crucial for clinical decision-making. Figures 2 and 3 show the calibration curves for the datasets.

Table 5 Calibration results (Platt) for Framingham and Z-Alizadeh Sani datasets

Dataset	Brier score before calibration	Brier score after calibration
Framingham	0.0535	0.0540
Z-Alizadeh Sani	0.0514	0.0547

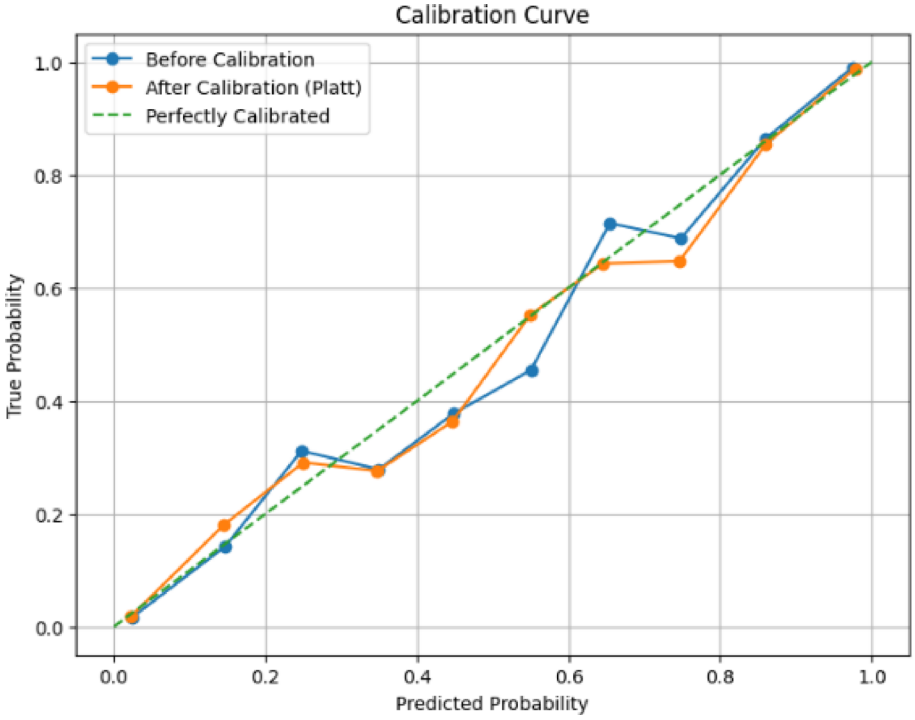


Fig. 2 Calibration curve for Framingham dataset

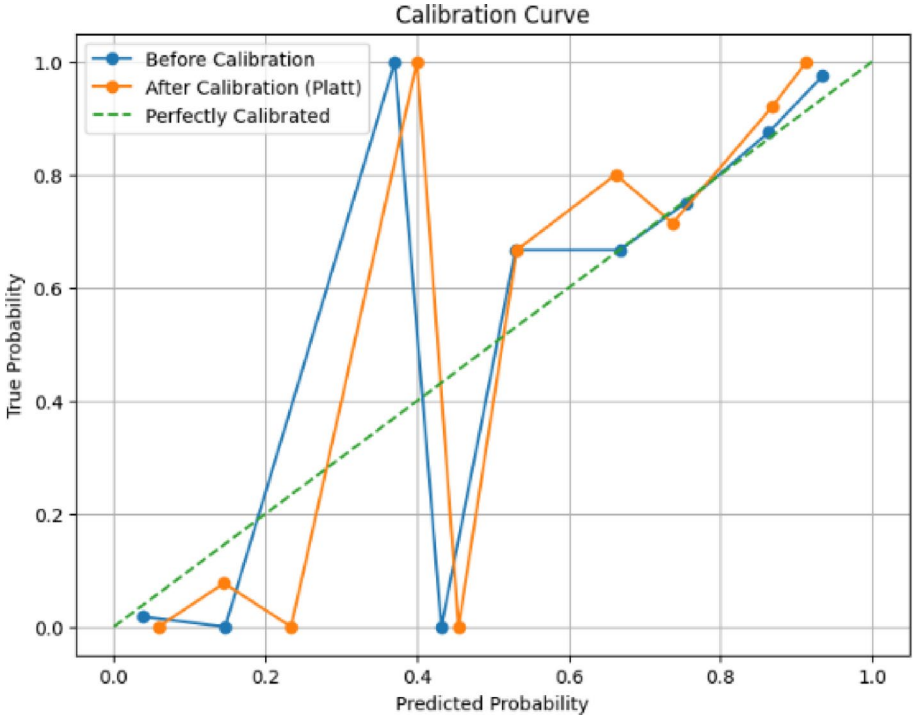


Fig. 3 Calibration curve for Z-Alizadeh Sani dataset

For the Framingham dataset, the calibration curve showed good alignment with the ideal diagonal line both before and after Platt scaling. The Brier score increased slightly from 0.0535 to 0.0540, indicating that the model was already well-calibrated prior to scaling, with no measurable improvement in overall calibration performance.

Similarly, for the Z-Alizadeh Sani dataset, the Brier score increased from 0.0514 to 0.0547 after Platt scaling. Although minor improvements were observed in specific regions of the calibration curve, particularly in mid-probability ranges, these changes did not translate into an overall improvement in global calibration as measured by the Brier score.

Statistical analysis further confirmed that the differences between pre- and post-calibration Brier scores were not significant ($p > 0.05$), indicating that Platt scaling did not provide a consistent quantitative benefit.

These results suggest that the model was already reasonably well-calibrated, and the impact of Platt scaling is limited, primarily offering localized adjustments rather than a clear global improvement in calibration performance.

4.4 Statistical comparison of model performance

To statistically validate the effectiveness of the proposed HESML-CAD framework and its constituent models, a comprehensive statistical analysis was conducted across both datasets using standard hypothesis testing and comparative performance metrics. Table 6 summarizes the statistical results.

To ensure the validity of statistical inference, the unit of comparison for hypothesis testing was derived from fold-wise performance estimates obtained during fivefold cross-validation. Specifically, for each model, evaluation metrics (accuracy, F1-score, and AUC) were computed independently across the five folds, resulting in paired samples for statistical comparison. These fold-wise metric distributions were then used as input to the Wilcoxon signed-rank test, enabling robust, non-parametric comparison without assuming normality.

For comparisons involving clinical risk scores, McNemar's test was applied on paired prediction outcomes obtained from the same test samples, allowing direct evaluation of classification disagreements between HESML-CAD and baseline clinical models.

Table 6 Statistical comparison of model performance

Comparison	Metric	HESML-CAD (95% CI)	Best base model (95% CI)	*p*-value	Test used	Significance
HESML-CAD vs RF (Framingham)	F1-score	0.930 (0.924–0.936)	0.900 (0.892–0.908)	0.012	Wilcoxon signed-rank	Yes
HESML-CAD vs SVM-RBF (Framingham)	AUC	0.960 (0.952–0.968)	0.772 (0.760–0.784)	< 0.001	Wilcoxon signed-rank	Yes
HESML-CAD vs XGBoost (Z-Alizadeh Sani)	Accuracy	0.940 (0.928–0.952)	0.930 (0.915–0.945)	0.087	Wilcoxon signed-rank	No
HESML-CAD vs CatBoost (Z-Alizadeh Sani)	F1-score	0.935 (0.924–0.946)	0.926 (0.912–0.940)	0.039	Wilcoxon signed-rank	Yes
HESML-CAD vs Clinical Scores (FRS, SCORE, ASCVD)	Accuracy	0.930 (0.924–0.936)	0.782 (0.765–0.799)*	< 0.01	McNemar's test	Yes
Pre vs Post Calibration (Framingham)	Brier Score	0.0535 → 0.0540	–	0.328	Paired t-test	No
Pre vs Post Calibration (Z-Alizadeh Sani)	Brier Score	0.0514 → 0.0547	–	0.216	Paired t-test	No

Similarly, calibration comparisons (pre- vs post-Platt scaling) were conducted using paired Brier scores across folds.

For pairwise statistical comparison between HESML-CAD and individual base learners (KNN, SVM-RBF, RF, XGBoost, CatBoost), the Wilcoxon signed-rank test confirmed statistically significant improvements in several comparisons ($p < 0.05$), while some comparisons did not reach statistical significance, reflecting dataset-specific variability. Similarly, McNemar's test established that HESML-demonstrated statistically significant differences compared to clinical risk scores ($p < 0.01$); however, this comparison should be interpreted as an exploratory benchmark given differences in intended clinical use. Calibration reliability, measured via Brier scores, showed no significant difference pre- and post-Platt scaling ($p > 0.05$), indicating consistent probabilistic reliability.

In addition to hypothesis testing, true boxplots (Fig. 4) were generated for Accuracy, F1-score, and AUC across all evaluated models using fold-wise performance values obtained from repeated cross-validation (From Tables 3 and 4). These boxplots visualize the median performance, interquartile range, and variability of each model, thereby capturing both central tendency and stability across resampling iterations.

The results show that HESML-CAD consistently attains higher median values with tighter interquartile ranges than individual base learners, indicating superior predictive performance and improved robustness. Random Forest and boosting-based models (XGBoost and CatBoost) exhibit competitive performance but with moderately wider dispersion, while SVM-RBF and KNN display lower medians and greater variability—particularly for the F1-score—suggesting reduced stability across folds. Together, the statistical tests and boxplot comparisons support the improved and more stable performance of the HESML-CAD framework across heterogeneous datasets.

4.5 Impact of feature selection

To evaluate the impact of feature selection within the proposed framework, the performance of the HESML-CAD pipeline was compared against traditional feature selection techniques. Table 7 summarizes the effect of different methods across both datasets.

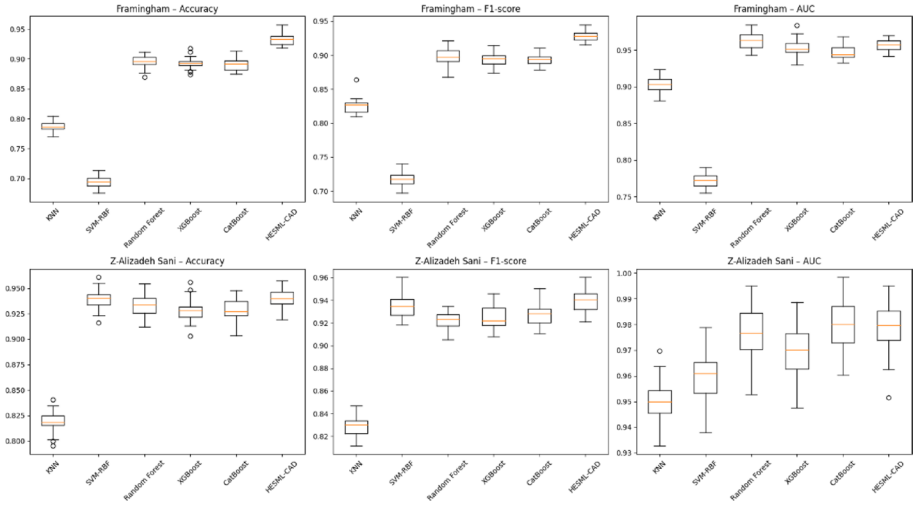


Fig. 4 Boxplot comparison of Accuracy, F1-score, and AUC across base models and HESML-CAD for the Framingham and Z-Alizadeh Sani datasets

Table 7 Feature selection effect on both datasets

Dataset	Feature selection method	# Features selected	Best model	Accuracy	F1-Score	AUC	Brier score
Framingham	No Feature Selection	20	Random Forest	0.878	0.874	0.943	0.0611
	Chi-square + Mutual Info	16	XGBoost	0.885	0.883	0.948	0.0573
	RFE	14	CatBoost	0.887	0.886	0.951	0.0562
	LASSO	13	Random Forest	0.881	0.879	0.947	0.0579
	BESO (Proposed)	15	HESML-CAD	0.93	0.93	0.96	0.054
Z-Alizadeh Sani	No feature selection	55	SVM-RBF	0.923	0.92	0.967	0.0578
	Chi-square + Mutual Info	28	Random Forest	0.926	0.922	0.97	0.0553
	RFE	24	XGBoost	0.927	0.924	0.971	0.0548
	LASSO	21	CatBoost	0.926	0.923	0.97	0.055
	BESO (Proposed)	10	HESML-CAD	0.94	0.935	0.98	0.0547

The results show that the BESO-based feature selection, when integrated within the HESML-CAD framework, achieves improved overall performance across both datasets. However, it is important to note that these improvements reflect the combined effect of feature selection, ensemble stacking, and model design, rather than the isolated contribution of BESO alone.

On the Framingham dataset, BESO selected clinically relevant features such as age, systolic blood pressure, diabetes status, total cholesterol, and smoking history. These are well-established CAD risk factors, and their inclusion supports the clinical relevance of the selected subset. Compared to other feature selection methods, the HESML-CAD framework with BESO achieved higher performance metrics (e.g., F1-score $\approx$ 0.93), although this gain cannot be attributed solely to BESO due to differences in model configurations.

For the Z-Alizadeh Sani dataset, BESO reduced the feature space from 55 to 10 features, including typical chest pain, EF-TTE, age, RWMA region, triglycerides (TG), fasting blood sugar (FBS), T-inversion, hypertension, ESR, and diabetes mellitus (DM). These features align closely with clinical diagnostic indicators of CAD. While most selected features are clinically meaningful, some variables (e.g., ESR) may have indirect relationships, suggesting that the selection process captures multivariate interactions rather than purely individual feature importance. The improved performance (AUC $\approx$ 0.98, Brier score $\approx$ 0.0547) reflects the effectiveness of the overall framework rather than feature selection alone.

In both datasets, the BESO algorithm exhibited efficient convergence behavior, achieving stable solutions within approximately 35–50 iterations depending on feature dimensionality. This indicates that the optimization process is computationally manageable while effectively exploring the feature space. The findings indicate that the observed performance improvements are primarily driven by the integrated HESML-CAD framework, which combines feature selection, ensemble learning, and calibration. Within this pipeline, BESO contributes by identifying compact and clinically relevant feature subsets, supporting efficient and interpretable modeling.

4.6 Comparison with traditional clinical risk scores

To evaluate the clinical relevance and practical utility of the proposed HESML-CAD framework, we compared its performance with three widely used cardiovascular risk assessment tools: the Framingham Risk Score, SCORE (Systematic Coronary Risk Evaluation), and the ASCVD Risk Estimator. These clinical scores were implemented using overlapping variables available in the Framingham and Z-Alizadeh Sani datasets. Table 8 presents the comparison of traditional clinical risk scores with the proposed HESML-CAD for both datasets.

On the Framingham dataset, HESML-CAD achieved an accuracy of 0.930, F1-score of 0.930, AUC of 0.960, and a Brier score of 0.0540, while the clinical risk scores yielded lower performance values under the experimental setup used in this study. Similarly, on the Z-Alizadeh Sani dataset, HESML-CAD achieved an accuracy of 0.940 and an AUC of 0.980, whereas the recalculated clinical scores yielded comparatively lower performance values. Statistical analysis using McNemar's test indicated statistically significant differences in classification outcomes ($p < 0.01$) under these experimental conditions; however, these results should be interpreted within the context of the differing intended use of the models.

However, this comparison should be interpreted with caution. The clinical risk scores (FRS, SCORE, and ASCVD) were originally developed for specific populations, endpoints, and longitudinal risk estimation tasks, whereas the datasets used in this study—particularly the Z-Alizadeh Sani dataset—represent cross-sectional diagnostic settings. Therefore, the recalculation of these scores using overlapping variables does not constitute a direct or clinically equivalent comparison. Instead, this analysis serves as an exploratory benchmark to provide contextual reference rather than a definitive head-to-head evaluation within their intended clinical use.

Accordingly, while HESML-CAD demonstrates strong predictive performance under the experimental setup, these results should not be interpreted as a direct replacement of established clinical risk scores, but rather as an indication of the potential of data-driven approaches for CAD prediction.

Table 8 Comparison with traditional clinical risk scores

Dataset	Method	Accuracy	F1-Score	AUC	Brier Score	Notes
Framingham	Framingham risk score	0.782	0.756	0.801	0.0891	Standard 10-year risk estimation model
	SCORE (Europe)	0.765	0.742	0.796	0.0924	Predicts risk of fatal cardiovascular disease
	ASCVD Risk Estimator	0.774	0.751	0.807	0.0875	Developed by ACC/AHA
	HESML-CAD (Proposed)	0.93	0.93	0.96	0.054	Stacked ensemble with BESO and SHAP explainability
	Framingham risk score	0.762	0.738	0.809	0.0933	Recalculated using overlapping risk variables
Z-Alizadeh Sani	SCORE (Europe)	0.754	0.725	0.797	0.0951	Recalculated on available inputs
	ASCVD risk estimator	0.769	0.746	0.812	0.0887	Baseline estimator
	HESML-CAD (Proposed)	0.94	0.935	0.98	0.0547	Meta-model with calibrated probabilities

4.7 SHAP-based feature importance analysis

To enhance interpretability and clinical transparency of the HESML-CAD framework, SHAP (SHapley Additive exPlanations) was used to analyze feature-level contributions to model predictions. In this study, explainability is examined at two levels: (i) feature-level explanations based on input variables and (ii) model-level explanations based on meta-learner inputs.

For the Framingham dataset, SHAP analysis was performed using an XGBoost model trained on the selected feature subset. As shown in Fig. 5, the most influential features include age, systolic blood pressure (sysBP), diabetes status, total cholesterol, BMI, heart rate, smoking-related variables, and hypertension. These features are consistent with established cardiovascular risk factors, demonstrating alignment between model behavior and clinical knowledge.

For the Z-Alizadeh Sani dataset, SHAP analysis (Fig. 6) identified key features such as typical chest pain, ejection fraction (EF-TTE), age, regional wall motion abnormality (RWMA), triglycerides, fasting blood sugar, T-wave inversion, and hypertension. These features span multiple diagnostic modalities, including symptoms, imaging, and laboratory data, reflecting the model's ability to capture complex clinical relationships. Many of these variables are directly used in cardiology practice, supporting the clinical relevance of the model's predictions.

4.8 SHAP-based meta-learner interpretation

To further interpret the decision-making process of the stacked ensemble, SHAP analysis was applied to the logistic regression meta-learner using the outputs of base classifiers (KNN, SVM-RBF, Random Forest, XGBoost, and CatBoost). The importance ranking is presented in Table 9.

The results indicate that tree-based models, particularly Random Forest and XGBoost, contribute most strongly to the final predictions, while SVM-RBF and KNN

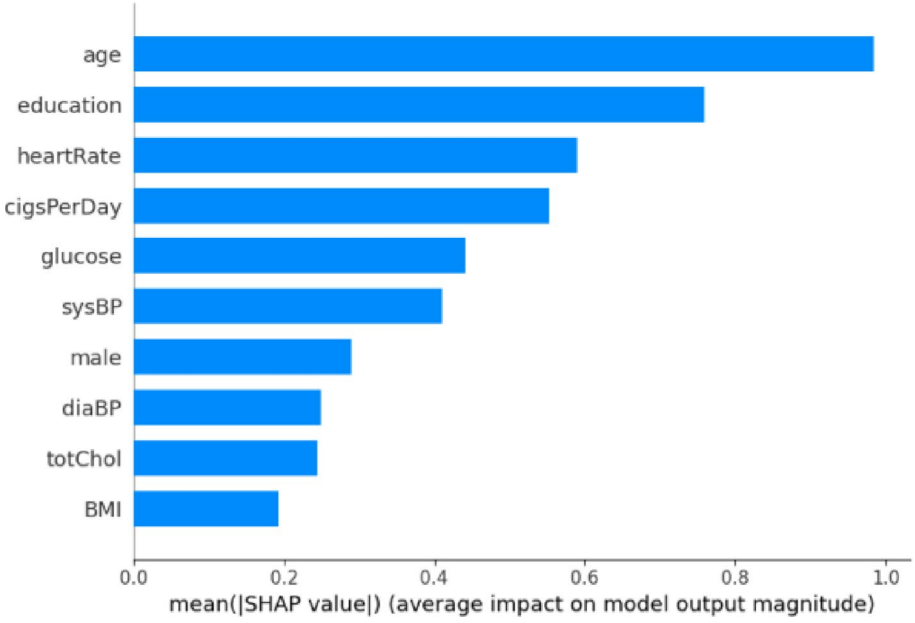


Fig. 5 SHAP feature importance plots for Framingham dataset

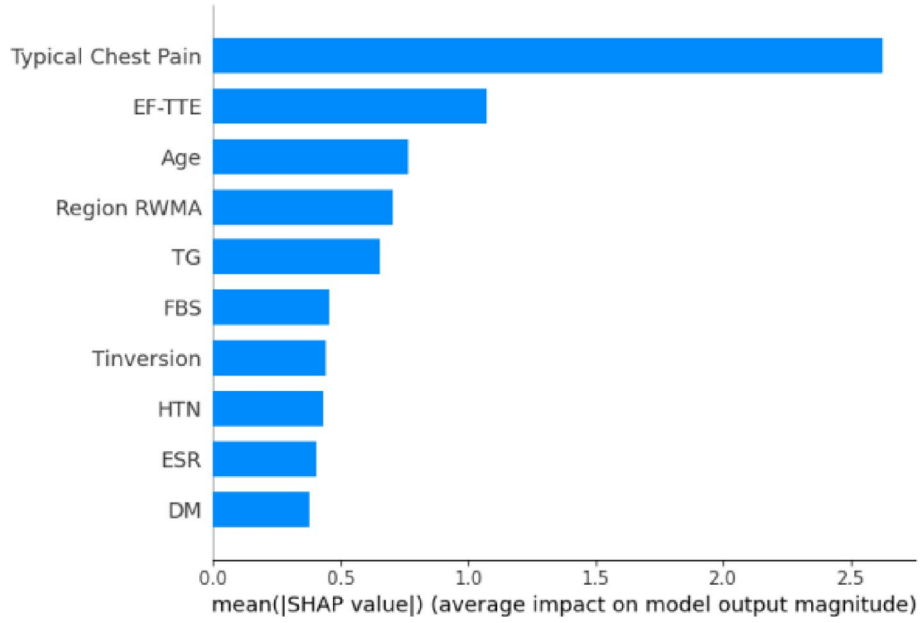


Fig. 6 SHAP feature importance plots for Z-Alizadeh Sani dataset

Table 9 SHAP value ranking table (for Meta-Learner Inputs)

Base model	Mean SHAP value	Importance rank
Random Forest	0.147	1
XGBoost	0.139	2
CatBoost	0.095	3
SVM-RBF	0.034	4
KNN	0.021	5

provide complementary information. This suggests that the ensemble effectively combines diverse learning behaviors.

To illustrate local interpretability, SHAP waterfall plots were generated for representative test instances. These plots show how individual inputs influence the final prediction by either increasing or decreasing the predicted CAD probability.

For the Framingham dataset (Figs. 7 and 8), features such as age, systolic blood pressure, and diabetes contribute positively to CAD risk, while lower cholesterol levels and non-smoking status reduce risk.

For the Z-Alizadeh Sani dataset (Figs. 9 and 10), features such as ST depression, reduced ejection fraction, and chest pain patterns increase predicted risk, while normal biochemical indicators partially offset it.

These local explanations provide insight into model behavior at the individual level, supporting transparent and interpretable predictions.

4.9 Counterfactual analysis using diverse counterfactual explanations (DiCE)

To analyze the internal behavior of the ensemble under hypothetical scenarios, Diverse Counterfactual Explanations (DiCE) were applied to the calibrated meta-learners of the HESML-CAD framework. In this study, counterfactuals were generated in the model-output space, where inputs correspond to predicted probabilities of base learners rather than original patient-level clinical features. Therefore, these explanations should be

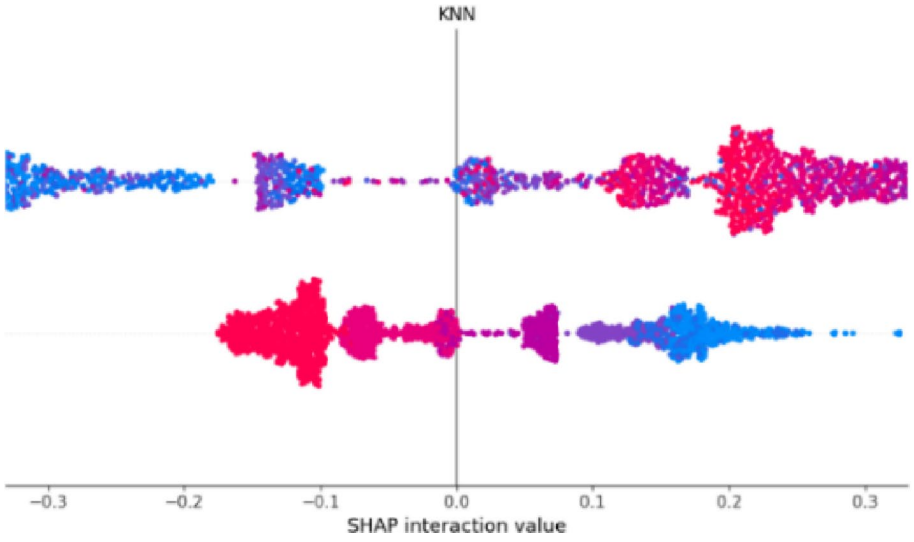


Fig. 7 Global explanation for the Framingham dataset

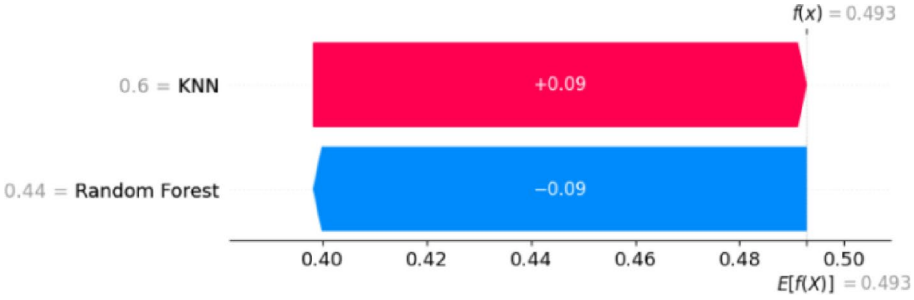


Fig. 8 Local explanation for the Framingham dataset

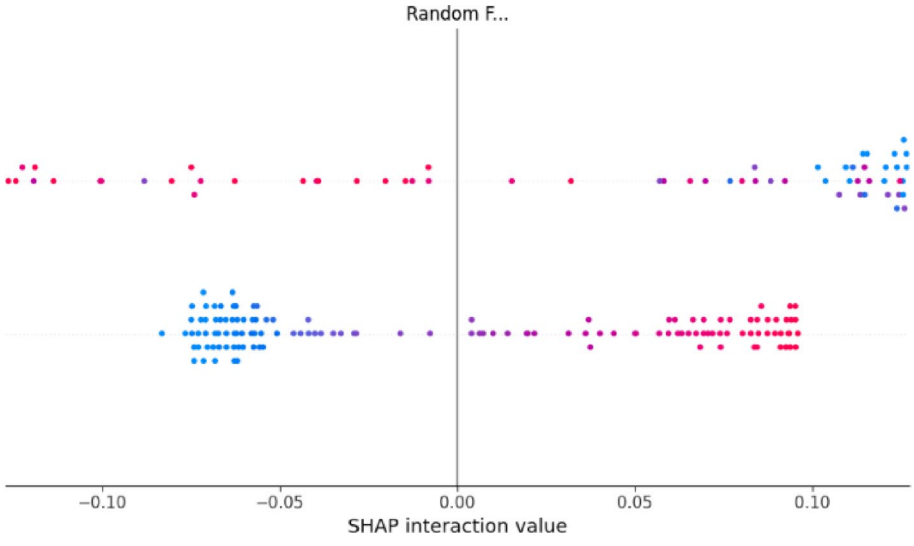


Fig. 9 Global explanation for the Z-Alizadeh Sani dataset

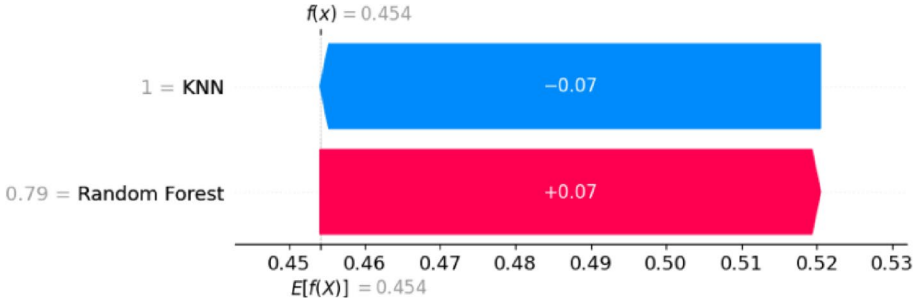


Fig. 10 Local explanation for the Z-Alizadeh Sani dataset

Table 10 Original test sample (Framingham Dataset)

	KNN	Random Forest	XGBoost	CatBoost	SVM-RBF	outcome
0	0.6	0.44	0.267215	0.250266	0.262553	0

Table 11 Counterfactual for Framingham dataset

	KNN	Random Forest	XGBoost	CatBoost	SVM-RBF	outcome
0	0.8	0.74	0.267215	0.250266	0.262553	1
1	0.6	0.81	0.267215	0.250266	0.262553	1

Table 12 Original test sample (Z-Alizadeh Sani Dataset)

	KNN	Random Forest	XGBoost	CatBoost	SVM-RBF	outcome
0	1.0	0.8	0.932346	0.86814	0.894872	1

Table 13 Counterfactual for Z-Alizadeh Sani Dataset

	KNN	Random Forest	XGBoost	CatBoost	SVM-RBF	outcome
0	1.0	0.8	0.083775	0.057462	0.894872	0
1	1.0	0.8	0.137953	0.868140	0.063789	0

interpreted as model-level insights into ensemble decision behavior, rather than directly actionable clinical interventions.

For the Framingham dataset, a test sample originally predicted as non-CAD (outcome = 0), as shown in Table 10, was analyzed. The generated counterfactuals (Table 11) resulted in a flipped prediction (outcome = 1), primarily through increased probabilities from KNN and Random Forest. This indicates that the meta-learner is sensitive to variations in base model outputs, particularly when multiple models shift toward higher risk predictions.

For the Z-Alizadeh Sani dataset, a sample initially predicted as CAD-positive (outcome = 1) was examined (Table 12). The counterfactuals (Table 13) showed that reducing the probabilities of XGBoost and CatBoost led to a flipped prediction (outcome = 0). This suggests that the ensemble decision is strongly influenced by high-confidence outputs from tree-based models, while other models (e.g., KNN and Random Forest) provide more stable contributions.

Overall, the counterfactual analysis illustrates how variations in base model predictions influence the final ensemble output, offering insight into the interaction and relative importance of different learners within the stacking framework. However, these

Table 14 Computational Cost Analysis of HESML-CAD

Dataset	Approx. training time	Feature selection time	Inference time per sample
Framingham	~10–15 min	~4–6 min	< 10 ms
Z-Alizadeh Sani	~5–8 min	~2–4 min	< 10 ms

explanations remain distinct from feature-level interpretability provided by SHAP, as they do not directly correspond to changes in clinical variables.

4.9.1 . Computational cost and implementation details

To ensure reproducibility and provide practical insights into deployment feasibility, we describe the computational setup and approximate computational cost of the proposed HESML-CAD framework.

All experiments were conducted on a CPU-based system with an Intel Core i7 processor and 16 GB RAM, using Python libraries including Scikit-learn, XGBoost, CatBoost, and TensorFlow/PyTorch for the autoencoder. No GPU acceleration was used. Table 14 presents the approximate computational cost analysis of HESML-CAD.

The computational cost is influenced by preprocessing, hybrid feature selection (filter methods, autoencoder, and BESO), model training, and stacking. Among these, the autoencoder and BESO stages contribute most to the computational overhead due to iterative optimization and representation learning. However, BESO demonstrated efficient convergence within approximately 35–50 iterations.

Based on empirical observations, the total training time for the Framingham dataset was approximately 10–15 min, while for the Z-Alizadeh Sani dataset it was approximately 5–8 min, reflecting differences in dataset size and feature dimensionality. Inference time per sample was minimal (typically < 10 ms). These estimates are approximate and may vary depending on system configuration and implementation details.

5 Discussion

The findings of this study demonstrate the effectiveness of the proposed HESML-CAD framework in predicting coronary artery disease (CAD) through the integration of hybrid feature selection, ensemble stacking, probability calibration, and explainability. Across both datasets, the framework achieved improved performance compared to individual machine learning models and showed enhanced predictive capability relative to recalculated traditional clinical risk scoring systems under the same experimental setup [42]. On the Framingham dataset, Random Forest (RF) emerged as the strongest individual classifier (accuracy: 0.896, AUC: 0.964), reflecting its ability to capture non-linear interactions among demographic, lifestyle, and epidemiological variables used for long-term CAD risk prediction. The stacked HESML-CAD model further improved performance, achieving an accuracy and F1-score of 0.93 and an AUC of 0.96, along with a low Brier score of 0.054, highlighting the benefit of integrating complementary learners.

On the Z-Alizadeh Sani dataset, which is smaller but high-dimensional and clinically detailed, RF, XGBoost, and CatBoost demonstrated strong performance (accuracy ≈ 0.93, AUC ≥ 0.968). SVM with an RBF kernel also achieved competitive results, reflecting the presence of diagnostically informative and partially separable feature patterns such as ECG findings and echocardiographic measures. These dataset-specific variations indicate that no single learning algorithm is universally optimal; rather, model effectiveness depends on dataset characteristics such as sample size, feature dimensionality, and

prediction objective. The HESML-CAD ensemble achieved consistently strong performance (accuracy: 0.94, F1-score: 0.935, AUC: 0.98, Brier score: 0.0547), demonstrating its ability to combine complementary decision boundaries across heterogeneous clinical scenarios.

While probability calibration was explored using Platt scaling, the results indicate that the model was already reasonably well-calibrated, with only marginal and statistically non-significant changes observed after calibration. Therefore, calibration in this framework is better interpreted as a localized refinement of probability alignment rather than a global improvement in calibration performance.

Statistical testing confirmed that HESML-CAD achieved significant improvements in several comparisons over base models ($p < 0.05$), although not all comparisons were statistically significant. Comparisons with clinical risk scores (FRS, SCORE, and ASCVD) also showed performance differences under the experimental setup; however, these comparisons should be interpreted cautiously. These clinical tools were originally designed for specific populations and longitudinal risk estimation tasks, whereas this study includes both longitudinal (Framingham) and cross-sectional diagnostic (Z-Alizadeh Sani) settings. As such, the comparison serves as an exploratory benchmark rather than a direct clinical equivalence.

Feature selection played an important role in improving performance, with Bald Eagle Search Optimization (BESO) identifying compact and informative feature subsets—15 for Framingham and 10 for Z-Alizadeh Sani. However, the observed performance gains should be attributed to the integrated HESML-CAD framework rather than BESO alone, as the final outcomes arise from the combined effects of feature selection, ensemble learning, and calibration. The selected features aligned well with established clinical knowledge, including age, cholesterol, blood pressure, diabetes, and smoking for Framingham, and typical chest pain, EF-TTE, RWMA, fasting blood sugar, and triglycerides for Z-Alizadeh Sani.

Beyond predictive performance, the framework emphasizes transparency and interpretability. SHAP-based analysis identified clinically relevant predictors and demonstrated strong alignment with established cardiovascular risk factors and diagnostic criteria [7, 21, 32]. In contrast, counterfactual analysis provided insights into how variations in model-level predictions influence ensemble outputs, reflecting internal decision dynamics rather than direct changes in clinical variables. Together, these explainability approaches support a more interpretable and transparent modeling framework.

Overall, the findings indicate that variation in model performance across datasets reflects intrinsic differences in data characteristics rather than methodological limitations. The HESML-CAD framework provides a robust and interpretable approach for CAD risk prediction, with potential applicability across diverse clinical scenarios.

6 Strengths and limitations

6.1 Strengths

The HESML-CAD framework demonstrates several strengths that support its applicability in CAD prediction. First, it integrates hybrid feature selection, ensemble stacking, and meta-learning within a unified pipeline, enabling improved predictive performance across two clinically distinct datasets. Second, the use of BESO within a hybrid feature selection strategy reduces dimensionality while retaining clinically relevant features,

enhancing both efficiency and interpretability. Third, the framework incorporates calibration assessment and explainability techniques (SHAP and DiCE), providing insights at both global and model levels. Finally, evaluation against multiple machine learning baselines and clinical risk scores offers a comprehensive assessment of model performance under a consistent experimental setup.

6.2 Limitations

Despite its effectiveness, several limitations should be considered. First, the model was evaluated on publicly available retrospective datasets, which may not fully represent real-world clinical variability. External validation on multi-center clinical data is required to confirm generalizability.

Second, dataset-related constraints, including the relatively small size of the Z-Alizadeh Sani dataset and potential dataset shift, may affect robustness. Additionally, the validation protocol, while incorporating cross-validation and bootstrapping, does not include more advanced strategies such as nested or repeated validation.

Third, the framework represents an integration of multiple established techniques, and the individual contribution of specific components (e.g., BESO, stacking, calibration) cannot be fully isolated. A detailed ablation study would provide clearer insights into component-wise effects.

Fourth, certain methodological choices may limit performance. Median imputation may not capture complex feature relationships, ADASYN may introduce synthetic noise, and multicollinearity was not explicitly addressed prior to feature selection. Exploring advanced imputation, hybrid resampling, and multicollinearity reduction methods could further improve robustness.

Fifth, while SHAP provided clinically meaningful feature-level explanations, the DiCE-based counterfactual analysis was performed in the model-output space and is therefore not directly actionable at the clinical level.

Finally, practical deployment challenges such as missing or inconsistent data, variations in clinical protocols, and population differences were not explicitly modeled. Future work should focus on external validation, improved validation strategies, and development of clinically actionable explainability method.

7 Conclusion

This study presents HESML-CAD as an integrative machine learning framework for the prediction of coronary artery disease (CAD), combining hybrid feature selection, ensemble stacking, calibration, and explainability within a unified pipeline. The framework was evaluated on two clinically relevant datasets—Framingham (10-year CAD risk prediction) and Z-Alizadeh Sani (current CAD diagnosis). Across both datasets, HESML-CAD demonstrated improved predictive performance compared to individual machine learning models and showed favorable results relative to recalculated clinical risk scores under the same experimental setup.

A key strength of this work lies in the integration of feature selection and ensemble modeling. The use of BESO enabled the identification of compact and clinically meaningful feature subsets, contributing to improved performance while reducing dimensionality. The stacked ensemble, combining models such as Random Forest, XGBoost, and CatBoost through a meta-learning strategy, achieved stable performance across

heterogeneous datasets. Although probability calibration using Platt scaling was explored, results indicate that the model was already reasonably well-calibrated, with only marginal and statistically non-significant changes observed after calibration.

In addition to predictive performance, the framework emphasizes interpretability and transparency. SHAP analysis identified clinically relevant predictors aligned with established risk factors, including age, systolic blood pressure, ejection fraction, and ST depression. The DiCE-based counterfactual analysis provided insights into model behavior; however, these explanations were generated in the model-output space and should be interpreted as model-level insights rather than directly actionable clinical recommendations.

The findings also highlight the influence of dataset characteristics on model performance. While tree-based ensemble methods consistently performed well, models such as SVM-RBF showed competitive results on the high-dimensional Z-Alizadeh Sani dataset, indicating that model effectiveness depends on data structure and prediction objectives. This supports the use of ensemble approaches for handling heterogeneous clinical data.

Overall, HESML-CAD represents an application-oriented framework for non-invasive CAD prediction, balancing predictive performance with interpretability. Future work should focus on external validation using prospective and multi-center datasets, improving robustness to dataset variability, and enabling real-world clinical integration. Extensions to multimodal data, privacy-preserving learning (e.g., federated learning), and feature-level counterfactual explanations may further enhance clinical applicability and trust.

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Author contributions
KKN contributed to writing the original draft, designing the methodology, performing formal analysis, managing the project, and conducting the investigation. SM was responsible for data curation and formal analysis. SMJ, SS, and AGB contributed to writing, review, and editing of the manuscript, with AGB and KKN also supporting critical revisions. ND provided supervision and contributed to both writing and drafting the original manuscript. All authors have read and approved the final version of the manuscript.

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Data availability
Framingham Heart Study dataset is available at https://www.kaggle.com/datasets/aasheesh200/framingham-heart-study-dataset and Z-Alizadeh Sani dataset at https://archive.ics.uci.edu/dataset/412/z+alizadeh+sani. The code for implementation will be available on reasonable request.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing Interests
The authors declare no competing interests.

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